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Program: Master of Business Administration (MBA)

Semester: IV

Course Code: DADS404

Course Name: Data Scraping

Assignment Set – 1

Q1. What factors should you consider when identifying a source for data scraping? [10 Marks]

When you are getting ready to write some code to scrape a website it is an idea to pick the right source first. This helps you avoid wasting your time and getting into trouble with the law. There are a things you need to think about.

First you need to make sure it is okay to scrape the website. You should check the websites terms of service and the robots.txt file. The robots.txt file tells you which parts of the website you are allowed to visit with your scraper. Even though it is not a law it is an idea to follow the rules. If you do not the website owner might have a case against you if they decide to take you to court. The law is not very clear on this issue. For example a court in the United States made a decision in 2022 about a case called hiQ v. LinkedIn. The court said that scraping data that's available to the public is different from accessing data that is protected. To be safe you should respect the websites rules. Use their official API if they have one.

Next you need to think about the quality and relevance of the data. Does the website have the information you need? Is the data up to date. How often is it updated? If you are working on a project that compares prices you need data that's current. If the website only updates its data a week it is not a good source for your project. You also need to think about how the data's formatted. If the website uses formats on different pages your scraper will need to be able to handle that. This can make your scraper more complicated and prone to errors.

The technology used by the website is also important. If the website uses HTML it is easy to scrape. You can use tools like requests and BeautifulSoup.. If the website uses JavaScript to load its content you need to use a headless browser, like Selenium or Playwright. This is slower. Uses more memory and it can break if the website changes its code.

You also need to think about how data you need to scrape and how often you need to do it. If you only need to scrape an amount of data once it is not a big deal.. If you need to scrape a lot of data every day you need to plan carefully. You need to think about how you will store the data and how you will avoid overloading the websites server. You do not want to get banned from the website.

Some websites have defenses to prevent scraping. They might use CAPTCHAs. Limit how many requests you can make. They might also use JavaScript to detect bots. You should try scraping a sample of data to see what defenses the website has.. Always check if the website has an official API. APIs are designed for scraping. They make it easy to get the data you need. You should only scrape the websites HTML if they do not have an API.

- * You should check the websites terms of service and robots.txt file.
- * You should think about the quality and relevance of the data.
- * You should consider the technology used by the website.
- * You should plan carefully if you need to scrape a lot of data.

Q2. Why are Wikipedia pages a preferred source for data scraping? Write steps to scrap data from a Wikipedia page using Python library BeautifulSoup. [5+5 = 10 Marks]

Wikipedia is a source for many practical reasons.

The content on Wikipedia is released under a Creative Commons licence.

This makes reusing the content straightforward long as you give proper credit.

The websites HTML code is clean and well-structured.

Tables on Wikipedia have a style called "wikitable".

Headings are marked with h2 and h3 tags.

Paragraphs are in p elements.

You do not need JavaScript to view the site.

The site does not block automated requests.

Wikipedia covers a range of topics.

It provides tables on subjects like country demographics and chemical elements.

Steps to Extract Data from Wikipedia

Step 1: Install dependencies

Run "pip install requests beautifulsoup4" in your terminal.

If you want to structure your data you can also install pandas.

Step 2: Import libraries

In your Python script add "import requests" and "from bs4 import BeautifulSoup".

Step 3: Fetch the page

Use "response = requests.get(url)" where url's the Wikipedia page URL.

Make sure the response status code is 200.

Step 4: Parse the HTML

Create a BeautifulSoup object: "soup = BeautifulSoup(response.text, 'html.parser')".

This converts the HTML into a format you can navigate.

Step 5: Locate the target elements

For a table: "table = soup.find('table' {'class: 'wikitable'})".

For all headings: "headings = soup.find_all('h2')".

You can also use CSS selectors.

Step 6: Extract and clean text

Go through each row and cell and use `get_text(strip=True)` to clean the text.

Build a list or a dictionary with the cleaned text.

Step 7: Store the output

Write to a CSV file using Python's csv module.

Create a pandas DataFrame and save it to a CSV file.

The result is a dataset ready, for analysis.

Q3. What are the advantages and disadvantages of API-based Scraping? [5+5 = 10 Marks]

When we use a platform's interface to collect data we get really good quality information. This is because the data comes to us in a format like JSON with clear names for each field. We do not have to search through HTML or deal with pages that are laid out differently.

The good things about using this method are many. For one the data is very reliable. This is because the platform's interface is versioned, which means they tell us about any changes and keep the old version working for a while. This is not like HTML, which can change suddenly without warning. Using the platform's interface also makes our legal position clearer because we are using the platform in a way that they have explicitly allowed.

Another good thing is that this method is more efficient. When we make a request we get the information we asked for instead of having to download the entire page.

However there are also some not good things. One problem is that the platform limits how requests we can make in a certain amount of time. This means that collecting a lot of data can take a time whereas scraping the website might be faster.

We also have to consider what data the platform makes available. The interface might not give us access to everything we can see on the website.

Then there is the issue of authentication, which can be complicated. We need to get keys manage tokens and handle other credentials, which can be a lot of work.

Lastly using the platforms interface can be expensive. For example Twitter, which is now called X started charging for its interface in 2023. This made it too expensive for many researchers. Other platforms have also started charging or increased their prices.

Advantages of API-based data collection include:

- * Data quality is high
- * Reliability is strong
- * Legal standing is
- * Efficiency is better

Disadvantages of API-based data collection include:

- * Rate limits restrict the volume of requests
- * Data scope is constrained
- * Authentication overhead is complicated
- * Cost is a growing concern and API-based data collection can be expensive like, in the case of Twitter, X.

Assignment Set – 2

Q4. Why is scraping tweets useful for data analysis? Explain the process of collecting tweets using an API from X. [5+5 = 10 Marks]

X, which used to be called Twitter produces an amount of public text data that people share their opinions on. This data covers every topic and is useful for analyzing public opinion.

- * It helps track what people think about a brand in time.
- * A sudden increase in comments can warn of a product problem or a public relations issue.
- * During events like elections, natural disasters or new product launches the number of posts on X changes minute by minute.

Financial experts have found that the way people feel about a company on X can affect its stock price in the term.

Researchers use X data to study how information spreads how people become polarized and how people behave when it comes to their health.

To collect data from X you need to apply for developer access through the X Developer Portal.

- * You describe why you want to use Xs data.

- * If approved you get credentials like a Bearer Token, API key, API secret and access tokens.

The X API has tools.

- * The recent search tool finds posts from the seven days that match what you're looking for.
- * The filtered stream tool gives you posts in time that fit certain rules.
- * The user timeline tool shows all posts from an account.

To use Xs API in Python you need to include your Bearer Token in the HTTP Authorization header.

Some libraries, like tweepy make it easier to authenticate and handle pagination.

- * You build a query with keywords, dates and languages.
- * You choose what data you want back like the post text, when it was posted and how many people interacted with it.
- * The data comes back in pages as JSON.
- * You use the "next_token" to get the page and pause, between requests so you don't hit the rate limit.
- * Usually people save the collected data in a CSV file or database for analysis.

Q5. Explain how data wrangling improves the quality of data with examples. [10 Marks]

Data wrangling is the process of making raw data clean and useful for analysis. It is really important because data scientists spend a lot of time 60 to 80 per cent getting the data ready instead of actually analyzing it. Each step in the data wrangling process fixes a problem with the data.

* **Missing value treatment.** Most of the time real datasets are not complete. For example a customer database with 50,000 records might have 3,000 phone numbers and 800 missing email addresses. There are a ways to deal with this such as deleting the incomplete records filling in the missing values with the average or median using a special model to predict the missing values or adding a special flag to show that the value was missing. In a dataset about housing prices filling in the square footage" values with the median value for the neighborhood helps to keep those records for analysis instead of throwing them away.

* **Duplicate removal.** Sometimes there are records in the data because of system errors, repeated form submissions or combining different data sources. If a customer appears three times their total purchase amount will be counted three times. Finding duplicates can be simple like looking for matches or more complicated like recognizing that "Jon Smith" and "Jonathan Smith" at the same postal code are the same person.

* **Data type correction.** If a date column is stored as text it cannot be used for date calculations or filtering. If a postal code is stored as a number it might lose its leading zeros. Fixing the data types is necessary for calculations.

* **Standardisation.** If a country field has versions of the same country like "US" "United States" "U.S.A." and "USA" it can make the data look more complicated than it is. Mapping all these versions to a standard value helps to simplify the data. For features scaling them to a common range ensures that no single feature dominates the analysis just because of its unit of measurement.

* **Outlier handling.** If a salary column has values between 30,000 and 150,000 but one entry is 99,999,999 it is probably an error. Using methods like z-scores or the interquartile range can identify these outliers and then the analyst can decide whether to correct them cap them at a plausible maximum or remove them. Each of these steps makes the dataset more reliable for analysis. Data wrangling is important for data analysis because it makes sure that the data is clean and useful. Data scientists use data wrangling to prepare the data for analysis. It is a crucial step, in the process.

Q6. Discuss the importance of using dplyr for preprocessing raw data. [10 Marks]

The dplyr package is a part of the ecosystem for R. It makes working with data a lot easier. This is because it has a way of doing things that is easy to understand.

The dplyr package is important when we are getting our data ready to use. It has three things about it: it can do a lot of things it works fast and we can use it with other things.

The package is built around five things we can do with our data.

* We can choose which columns we want to use with the function. For example if we have a lot of data with 80 columns but we only need 5 of them we can use select to pick the ones we want like `select(name, email, amount, date)`.

* We can filter our data to only include the rows we want. For instance if we want to see the records where the amount is more than 100 and the region is the West we can use `filter(amount > 100 region == "West")`.

* In the version of R we can do the same thing but it is harder to read and understand. It would look something like this: `df[df$amount > 100 & df$region == "West" ]`.

* We can also add columns or change the ones we have with the mutate function. For example if we want to add a column called profit we can do it like this: `mutate(profit = revenue. Cost)`.. If we want to get the month from a date we can do it like this: `mutate(month = lubridate::month(order_date))`.

* We can sort our data with the arrange function. If we want to see the records with the sales first we can use `arrange(desc(sales))`.

* We can use the summarise function with the group_by function to get a summary of our data. For example if we want to see the sales for each region we can do it like this: `group_by(region) |> summarise(avg = mean(sales) n =. )`).

One of the things about the dplyr package is that we can use the pipe operator to do a lot of things one after the other. This makes it easy to read and understand what we are doing with our data. It is like saying "take the data filter it then change it then summarise it".

The dplyr package is also very fast. This is because some of the code is written in C++. If we have a lot of data that does not fit in the memory of our computer we can use dbplyr to send the code to a database and get the results back. This way we can write the code regardless of where our data is.

There are some things we can do with the dplyr package.

- * We can rename columns with the function. This is useful if the names of the columns are not good.
- * We can remove duplicate rows with the function.
- * We can merge two datasets with the left_join function.
- * We can use the case_when function to make decisions based on conditions.
- * We can use the function to do the same thing to many columns at once.

All of these things together make the dplyr package very useful for getting our data ready to use. It makes it faster to write the code easier to check and more efficient to run. The dplyr package really helps us when we are working with data, in R. We can use the dplyr package to do a lot of things with our data.